5. Introduction of Programming and Software

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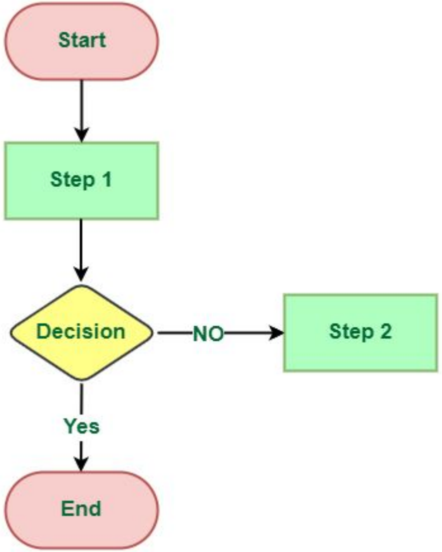
Content

1. Algorithm
2. Programming Language
3. Code
4. Program
5. Solve Problem and Create Program
6. Software
7. Some Software Application

1. Algorithm

- An algorithm is a step-by-step set of instructions to solve a problem or complete a task. It is a plan.
- An algorithm is the idea or the logic behind the solution. It's not written in a programming language yet; it can be in **plain English or a flowchart**. It's the "what" and "how" of the process.
- **Key Characteristics**: It's unambiguous, well-defined, and has a clear starting and ending point.



Algorithm, Pseudocode & Flowchart - 1

Algorithm & Flowchart to find the sum of two numbers

Algorithm

Step-1 Start

Step-2 Input first numbers say A

Step-3 Input second number say B

Step-4 SUM = A + B

Step-5 Display SUM

Step-6 Stop

Pseudocode

Begin

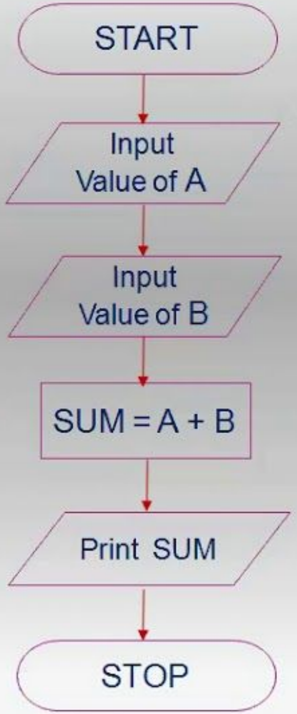
Input A

Input B

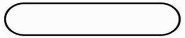
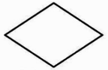
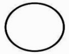
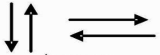
Compute SUM = A + B

Print SUM

End



Flowchart Symbol

Name	Symbol	Purpose
Terminal	 oval	Start /stop/begin/end
Input / output	 Parallelogram	Input/output of data
Process	 Rectangle	Any processing to be performed can be represented
Desicion box	 Diamond	Decision operations that determine which of the alternative paths to be followed
Connector	 Circle	Used to connect different parts of flowchart
Flow	 Arrows	Joins 2 symbols and also represents flow of execution
Pre defined process	 Douuble sided rectangle	Modules (or)subroutines specified else where
For loop symbol	 Hexagon	Shows initialization, condition and incrimination of loop variable.
Document	 Printout	Shows the data that is ready for printout

2. Programming Language

- A **programming language** is a formal language used to write programs.
- It is used to write instructions that a computer can understand and execute.
- It provides the syntax (grammar) and keywords (vocabulary) that a programmer uses to write code. Each language has its own strengths and is suited for different tasks (e.g., web development, data science, mobile apps).

2. Programming Language

Front-end web development
- JavaScript
- Elm
- TypeScript
- Cascading Style Sheets (CSS)

Back-end web development
- JavaScript
- TypeScript
- Python
- Go
- Ruby
- Scala

Mobile development
- Swift
- Java
- C#

Game development
- C#

Desktop applications
- Java
- Python
- JavaScript
- TypeScript
- Scala
- Go

Systems programming
- C#
- C++
- Go

3. Code

- Code (or source code) is the set of instructions written in a specific programming language that implements an algorithm.
- Code is the text file that a programmer creates. It is human-readable (for those who know the language) but cannot be directly executed by the computer's brain (the CPU).

```c
#include <stdlib.h>
#include <stdio.h>

int main (void) {
  printf("Hello, World!\n");
  exit(0);
}
```

Fig: Code in C language

4. Program

- A program is the final, runnable product after the code has been compiled or is being executed by an interpreter **(compiler/interpreter = translator)**. It is a sequence of machine code instructions that the computer's CPU can understand and execute directly.
- When we double-click on `chrome.exe` or `notepad.exe`, we are running a program.
- So, a Program is the executable file we run to perform a specific task.

5. Solve Problem and Create Program

5.1. Program - Grade Point Calculation (using C programming language: Code

```c
#include<stdio.h>
int main()
{
    float res, organic_pharm, organic_pharm_lab, inorganic_pharm, inorganic_pharm_lab,
    pharmacognosy, pharmacognosy_lab, basic_math, english, computer;

    printf("Enter the GPA of Organic pharmacy = "); scanf("%f", &organic_pharm);
    printf("Enter the GPA of Organic pharmacy Lab = "); scanf("%f", &organic_pharm_lab);
    printf("Enter the GPA of Inorganic pharmacy = "); scanf("%f", &inorganic_pharm);
    printf("Enter the GPA of Inorganic pharmacy Lab = "); scanf("%f", &inorganic_pharm_lab);
    printf("Enter the GPA of Pharmacognosy = "); scanf("%f", &pharmacognosy);
    printf("Enter the GPA of Pharmacognosy Lab = "); scanf("%f", &pharmacognosy_lab);
    printf("Enter the GPA of Basic math = "); scanf("%f", &basic_math);
    printf("Enter the GPA of English = "); scanf("%f", &english);
    printf("Enter the GPA of Computer Science = "); scanf("%f", &computer);

    res = (organic_pharm*3) + (organic_pharm_lab*1) +
    (inorganic_pharm*3) + (inorganic_pharm_lab*1) +
    (pharmacognosy*3) + (pharmacognosy_lab*1) +
    (basic_math*3) + (english*3) + (computer*3);

    res = res/21;
    printf("\nTotal CGPA of this semester = %.2f", res);
    printf("\n\n");
    return 0;
}
```

5. Solve Problem and Create Program

5.1. Program - Grade Point Calculation: Output(.exe file):

E:\SPRING_APRIL_26\a.exe

```
Enter the GPA of Organic pharmacy = 3.25
Enter the GPA of Organic pharmacy Lab = 3.50
Enter the GPA of Inorganic pharmacy = 4
Enter the GPA of Inorganic pharmacy Lab = 3.20
Enter the GPA of Pharmacognosy = 3
Enter the GPA of Pharmacognosy Lab = 4
Enter the GPA of Basic math = 3.25
Enter the GPA of English = 4
Enter the GPA of Computer Science = 4

Total CGPA of this semester = 3.58
```

6. Software

- Software is a collection of one or more programs, along with associated data and documentation, designed to fulfill a particular purpose for the user.
- So, Software is the entire suite of programs and resources that provide a full-featured application.
- For example, Microsoft Word is software. It consists of the main program (winword.exe), helper programs for spell check, printing, etc., and all its data files (templates, dictionaries). A video game, our mobile banking app, and the Windows operating system itself are all examples of software.

6. Software

Let's tie it all together with an analogy of building a house i.e. **Software**:

1. **Algorithm**: The architect's plan and blueprint for the house. *(The idea)*
2. **Programming Language**: The set of construction rules and terminology the builders will use (e.g., metric system, specific building codes). *(The rules of communication)*
3. **Code**: The detailed work orders and instructions written for the construction crew, based on the blueprint and using the correct terminology. *(The written instructions)*
4. **Compiler/Interpreter**: The general contractor who interprets the work orders and directs the crew to execute them. *(The translator/manager)*
5. **Program**: A single, finished component of the house, like the fully constructed roof or the installed plumbing system. *(A functional component)*
6. **Software**: The entire, finished house, complete with roof, plumbing, electrical systems, paint, and landscaping, ready for you to live in. *(The final product)*

8. Some Software Applications

END
of
chapter 5